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Collaborative working for large digitisation projects

Robin Yeates
London Borough of Barnet, Ilford, UK
Damon Guy
Slough Libraries, Arts and Information Services, Slough, UK

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Abstract

Purpose – To explore the effectiveness of large-scale consortia for disseminating local heritage via the web. To describe the creation of a large geographically based cultural heritage consortium in the South East of England and management lessons resulting from a major web site digitisation project. To encourage the improved sharing of experience amongst similar projects in the future.

Design/methodology/approach – A selective literature review seeking lessons on sustainable cultural heritage collaboration is presented. The composition of a consortium set-up with public lottery funding to build a cross-domain collaborative public web site containing cultural heritage materials from many local authorities and local partners is described. Practical experiences from the first three years of collaboration are presented in a manner loosely based on the European Union Digitisation Policies Benchmarking Model.

Findings – Staff in local government libraries, archives and museums have, until recently, lacked experience in major web site publishing and even in the basic digitisation of locally held heritage materials. Local governments in England have traditionally co-operated within the library or museum sectors, but not across the whole domain. New communication methods and a formally structured consortium have so far enabled the collaborative publication of a major web site allowing cross-searching of partner materials as well as access to individual sites. More importantly, staff have begun to understand how to proceed in a sustainable way to support the future development of more sophisticated digital primary and learning resources that are preserved for the future and yet accessible to many more people.

Practical implications – The paper notes the importance of shared values, common objectives and a practical approach to collaborative service delivery. It also highlights the challenges in situations where operational staff have multiple responsibilities and project staff may be available only while external short-term funding lasts. It is suggested that encouraging the documentation of operational experiences and sharing skills through a large-scale formal consortium can support sustainable development.

Originality/value – The consortium described hopes to form a major part of the future framework for digital cultural heritage services in the South East of England. This paper offers a rare, reflective contribution from experienced, operational service practitioners who have completed a three-five year digitisation programme offering free public access to materials that serve to support and disseminate local and regional identity.

Keywords Digital libraries, Cultural studies, England, Public libraries, Museums, Archiving

Paper type Case study

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1. Introduction

We provide a case study of the setting up of a new "smart" organisational structure, intended not to create economic value directly, but to build what might be called

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"cultural capital" by developing innovative ways to deliver learning resources provided predominantly by local governments.

Strong democracy, social capital, individual empowerment, sense of community and economic development opportunities are foci for assessment of the impact of emerging community networks and technology centres (O'Neil, 2002).

Holley (2004) has reviewed the global picture for heritage digitisation projects, including the Scottish Cultural Resources Access Network, SCRAN (www.scran.ac.uk), which was the largest UK trailblazer for collaboration to create digital learning resources. SCRAN received £15 m initially, using it to gather and publish on the web and offline a wide range of materials about Scotland that were distributed in collections in Scotland and beyond, to catalogue them using high quality metadata and to develop rights management principles and licences appropriate to such projects.

The UK Government's Department for Culture, Media and Sport (DCMS) set-up a Challenge Fund which, inter alia, identified the "pros and cons" of collaborative working within the UK museum domain (Resource (now MLA), 2002). Although work has been undertaken in consortium-based library management systems sharing (Froud, 1999), few cross-domain collaborative digitisation projects from the cultural sector have yet reported on such issues in the UK, but a significant increase in experience will have been gained as a result of the recent New Opportunities Fund (NOF) People's Network Content Creation programme (NOF-digitise, 2002) that began in 2001 and ended December 2004.

This paper highlights one of the consortia, the Sense of Place South East (SoPSE) formed as a result of a grant awarded by the government-backed, lottery-funded NOF (www.sopse.org.uk). At the outset, there was little experience of the creation and delivery of digital materials within the consortium. Neither was there much individual experience of working as part of a larger project. Yet the consortium has now created a formidable array of digital objects and learning materials, delivered via a single web site. Figure 1 shows the homepage of the SoPSE web site.

Taking the project as an example, the paper will articulate the steep learning curve involved in such types of public-sector digitisation projects.

2. Cross-domain collaboration

O'Neil (2002) suggests the need to plan evaluation carefully:

- state its purpose;
- involve stakeholders;
- involve participants in evaluation;
- use appropriate quantitative data;
- triangulate to corroborate results; and
- validate methods with experts.

SoPSE has not yet carried out this level of long-term evaluation, but is already reflecting on its experiences compared with those who are further advanced.

Three factors seem essential to create sustainable digital information and finding aid services:

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(1) a common vision;
(2) a collaborative institutional infrastructure; and
(3) adequate funding.

Evidence for this comes, for example, from California over the last ten years or so (Chandler, 2002). It is also necessary for cultural institutions to go beyond the provision of mere databases of disparate objects and intellectual items, to create compelling navigational and learning experiences for end-users and to provide appropriate contexts for use and learning (Nickerson, 2002). Nickerson, citing Sherwood (1998) at the Canadian Heritage Information Network suggests that co-operation between academia and museums "will make cultural heritage materials more accessible and meaningful, and these partnerships may also prove to be more economical and efficient".

Perry (2002), based on experience in Rochester, New York, has pointed out that not every approach to collaboration will be successful. Project complexity, the need for new focuses for innovation and the provision and development of new skills can threaten sustainability. He recommends central technical and production processes based on contractual business models, combined with decentralised intellectual and knowledge processes within small, autonomous work units. Project management authority must be combined with user integration and needs to be sufficiently strong to deliver just-in-time creation of historical contexts in educational products, by building a new hybrid knowledge base within the collaborative work unit.

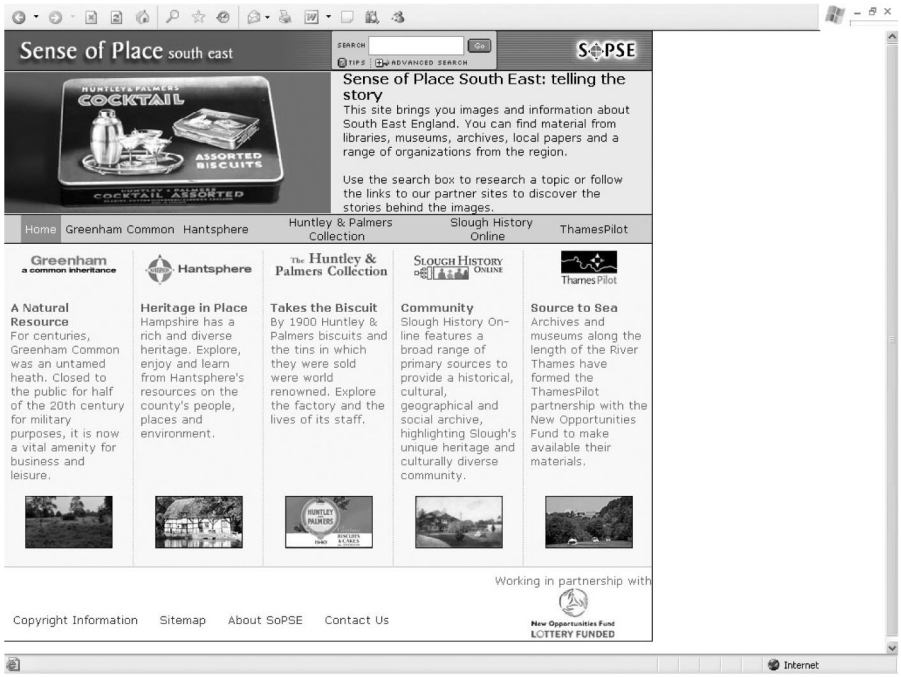

Figure 1.
Homepage of the SoPSE web site

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Taylor *et al.* (2001) have suggested that staff development for web service building is often best undertaken within an operational context, since future skills needed are still largely uncertain. These authors have also reviewed 20 UK case studies that "appeared to indicate that organisations should consider developing frameworks/standards/best practice guides in order to assist IT staff in web development projects". Such toolkits are becoming more common in UK higher education, and Resource, now the Museums, Libraries and Archives Council, developed links with this experience via the UKOLN/AHDS Technical Advisory Service for NOF-digitise. However, there is still a need for consortia such as SoPSE to develop their own knowledge-base.

The development of good practice guides is itself dependent on time for reflection and an appropriate knowledge support environment. The focus of large-scale digital content development programmes has thus far been on the quantity and quality of content developed. It might be expected that a stronger, more sustainable organisation will endure where good practice is strongly supported by shared manuals and knowledge-bases as well as locally developed guidelines and procedures.

3. The SoPSE baseline

The case study approach taken in this paper is loosely based on policy benchmarking work undertaken within the framework of the European Commission's eEurope Digitisation Initiative. This recognises that collaborative digitisation projects are complex, and that for comparative assessment to be useful it needs to be rigorous and open and include certain cross-project themes. The European Union's Digitisation Policies Benchmarking Model has identified themes including:

- management (objectives, workplan);
- human resources (available skills);
- funding (including sustainability);
- productivity (including the proportion of content that has been digitised);
- impact (added value);
- priorities (selection criteria for digitised materials); and
- technical aspects (appropriate technologies).

We use these broad themes to describe the SoPSE consortium and lessons learned so far, then we suggest how these lessons might be used to improve the assessment of collaborative digitisation projects, particularly those in the public cultural heritage domain.

A complementary approach might also use the partnership evaluation toolkit produced by Suffolk County Council (2002) and made available on the IDeA Knowledge web site. This focuses assessment on questions related to action, efficiency, inclusivity, learning and development, and performance management.

3.1 Management

The People's Network was intended as a project to connect all public libraries to the internet, as part of the government's commitment to give everyone in the UK the opportunity to get online. It was funded by the UK's National Lottery through NOF and managed by Resource: The Council for Museums, Archives and Libraries

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(now the Museums, Libraries and Archives Council). By the time of the Tavistock Institute project impact evaluation (Big Lottery Fund, 2004) in November 2004, 30,000 broadband connected terminals were in use at over 4,000 public libraries by over 16 per cent of the population over the age of 16. One of the three strands of the project was the Digitisation of Learning Materials Programme (also known as NOF-digitise), designed to produce new, unique content for the network. The overall aim of the People's Network is to offer opportunities to everyone, regardless of age or individual circumstances. An important objective of the NOF-digitise programme is to engage as many people in informal and formal learning as possible, by marrying together effectively an infrastructure of free network access provision, trained staff to assist users and appropriate freely available content.

Consortia, involving mainly local government bodies and their suppliers, bid for some £50 million funding in two stages. After the first stage expressions of interest, most selected projects were further combined, with encouragement by the funding body, into larger, more complex consortia. This process resulted in some 150 separate project web sites and 37 consortia gateways to new content, involving 500 content contributors (Anderson and Mitchell, 2002). However, the public may well judge the success of this work not by the quantity of web sites, but by their level of integration, quality of content and accessibility.

The SoPSE consortium brings together five organisations as main partners and a number of associate partners who are working locally on projects. The main partners are organisations leading a local partnership formed to develop digital content, shown in Table I.

The heritage themes covered add to the pool of knowledge about the South East region of the UK. All have natural synergies and parallels.

The SoPSE consortium works together with the following overall aims:

- Form an integrated multi-agency consortium affording wide access to source materials in the archive for the public.
- Create and share overall project and financial management.
- Adopt the same technical and delivery standards.
- Share some hardware and software resources in service delivery.
- Contract/purchase some goods and services as a consortium.
- Aim to manage all resources to achieve Best Value.
- Develop, share and adopt principles and practice for quality standards in digitisation.

The SoPSE consortium aims to create a unique, diverse and extensive resource through the effective use of ICT to provide new learning opportunities through the exploitation of a wide range of existing materials relating to the South East region of England.

The partners have stated:

> The major objective of this project is to enable the resources provided by associate partners to be more accessible. All partners seek to preserve information about published and archive material, culture and the environment. Additionally, the project will contribute to community facilities and promote lifelong learning and skills development. The resource will enhance the South East region's cultural heritage, sense of place and a feeling of local pride and community identity.

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Table I.
SoPSE themes and topics

Topic	Description	Outcome	Lead body
Hantsphere	The development of an innovative multimedia resource discovery database	One-stop access to the historical heritage of Hampshire utilising search engine and geographic information system technology to enable rapid and accurate information retrieval	Hampshire County Council Library Service
Huntley and Palmers Digitisation Project	The creation of a unique resource of information on the web from the Huntley and Palmers collection	The provision of free and user friendly access to a large and comprehensive collection of primary source material about a world famous and typically British industry	Reading Museum and Archive Service
ThamesPilot	The creation of a multimedia interface to local history for communities along the River Thames	A collection of material relating to the River Thames from source to sea including digitised images, sound and video	Royal Borough of Windsor and Maidenhead
Slough History Online	The creation of a series of resources and exhibition web pages to celebrate the history of the development of Slough	The development of an online cultural and educational resource that is a digital reflection of the social, economic, historical and geographical attributes of a unique town in the twenty-first century	Slough Borough Council, Libraries, Arts and Information Service
Greenham Common	The creation of an interactive lifelong learning web site for the presentation and interpretation of a selection of materials relating to Greenham Common from the period 1939 to the present	A web site using a selection of digitised historical, artistic, social, political, environmental and technical information about the WW2 and cold war use of Greenham Common including the women's peace protest movement	West Berkshire Council

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Each of the projects has its own business plan, mission and objectives, structured as recommended by NOF-digitise. They came together in a SoPSE business plan at stage two of the bidding process, and have planned detailed activities within a joint SoPSE framework. These business plans were a requirement of the original bid for the NOF-digitise grant. These have never been published, but they represented the way each project saw its operation and future management. In the second phase of the bidding process the newly formed consortium also submitted a revised business plan for the consortium which incorporated the partners' individual plans. These documents became the guiding document that each project worked to with respect to the demands made by NOF to meet the terms of the bid (copies of the Slough History Online Business Plan are available from Damon Guy).

As an example of the kind of associate partners in the project, ThamesPilot includes material from many of the local authorities along the river Thames from its source to the sea, as well as the River and Rowing Museum at Henley and the Company of Watermen and Lightermen in London – some 13 partners in all.

The SoPSE partnership was initially planned to operate until December 2003, but with responsibility for maintaining the site for at least a further three years. Work actually continued into 2005 with longer term consortia arrangements being currently under development.

3.2 Human resources

The consortium originally was:

- Very different from today! Key staff, in particular, notice their perspectives have changed vastly.
- Local government people. Little consideration had been given to including all stakeholders to maximise community benefits.
- Bidding for funds as disparate groups. The individual projects originally varied enormously in their approach to budgeting and priority setting, potentially competing for funds and overlapping in effort.
- Working to independent/diverse project goals. Ambitions to create high quality resources and services looked like producing incoherent outcomes for target audiences that might well threaten sustainability.
- Limited by lack of awareness and understanding, risking low quality services.
- Paying attention to funding issues rather than quality planning.

However, the fixed-term challenge funding suggested we could better exploit the diverse skills available to partners and build sustainable local capacity that might encourage regeneration of services rather than rely entirely on outsourcing to achieve quick fixes. We had:

- some individual technical experience;
- librarians, museum and archive people;
- technical experts in some subjects;
- loads of enthusiasm and contacts;

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- no experience of consortia; and
- no experience in large-scale digital projects.

We, therefore, planned to recruit new local project management staff, develop new types of digitisation and learning resource creation capacity within the consortium, adjusting existing management practices where necessary. We recognised this would sometimes lead to delays and frustrations, but in the end services should better meet local needs. Training and staff development, including consideration of appropriate staffing structures, would receive a new impetus as a result of the partnership working.

3.3 Funding

After reviewing the separate Stage 1 bids with external expert assistance from the academic sector, those assessing the NOF-digitise bids saw synergy in projects working together. It aimed for:

- something greater than the sum of the parts;
- sharing resources along the way; and
- maximising cost-efficiency and cost-effectiveness.

We would need to address the current lack of:

- existing financial management mechanisms for shared infrastructure;
- multi-authority joint planning for cultural heritage development work; and
- integrated funding for cross-domain work by museums, libraries, galleries and archives.

We also recognised the longer-term need to attract additional future partners to meet the most ambitious expectations of stakeholders.

3.4 Productivity

None of the partners had existing large volumes of relevant digitised content; nor did they operate suitable content management systems (CMS) that could be extended to achieve SoPSE objectives.

In selecting the material proposed to NOF for digitisation, partners took into account relevance to the individual themes, the potential audiences that each project had identified, sustainability, ensuring a balance of material, historical and local importance, and how these would contribute to the SoPSE Project. Quality, not quantity, was aimed for, with a capacity to extend the service according to needs. However, within the various projects, it would be necessary to digitise appropriately a variety of types of material including:

- scattered audio and video resources;
- lengthy runs of local newspapers;
- maps;
- artefacts;
- survey data;
- prints, drawings and paintings; and
- thematic selections of images and texts.

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There was a clear need to survey collections in further depth to plan content selection, but this also required skills and a dedicated project management architecture and infrastructure to be developed.

3.5 Impact

Although certain partners had experience of managing small cultural heritage web services, surveys of partners at the outset showed little evidence of transferable value in terms of guidelines, standards or skills relating to usability planning, preservation, metadata management or promotion of lifelong learning and social inclusion.

It was, therefore, expected that the SoPSE consortium might usefully address these issues on behalf of all partners, to match the overall aims and objectives and work towards the more measurable impact of cultural heritage services.

3.6 Priorities

We started with few explicit criteria to direct scarce resources to digitising the most appropriate materials. SoPSE was formed almost in parallel with the process of developing such criteria and the skills to generate them, since partners had been brought together only by the funding body.

The priority setting key task balanced exceptional opportunities for local authorities with risks. Consortium-wide working should provide new levels of quality, meeting "Best Value" aims (important for UK local government strategy and performance) and leveraging existing knowledge, although there were perceived risks in terms of imbalances in contributions and benefits to partners.

The Board of SoPSE carried out a comprehensive internal project risk assessment, resulting in a series of quality intervention points that the board could use to assess their progress in preventing problems. The assessments were published in an abridged form in 2002 on the NOF web site (the full assessment is currently available from Robin Yeates). During the course of the project, risk intervention by the board was seen as most important. It was a way of saving money as well as ensuring that the projects collectively focussed on the NOF-digitise guidelines and the individual project aims and objectives.

3.7 Technical aspects

Collaborative working on service delivery meant ICT staff and external advisers were involved in planning from the outset of the stage two bidding process. The Technical Group made a number of proposals, accepted by the board, that were included in the successful funding bid.

A key decision, made for technical, operational and financial reasons, was to offer services via a central server platform operated by Hampshire County Council. After surveys of partner facilities, no suitable open standards-based operational platform was found to exist, but the bandwidth and support required could be best made available at a location in Winchester. So, for reasons of operational ease and a supportive environment, a Microsoft server-based solution was provided by Hampshire County Council.

As recipients of the NOF grant we needed to ensure compliance with open protocols such as Simple Object Access Protocol (SOAP-www.w3.org/TR/soap/) and to pursue open scripting standards meeting World Wide Web Consortium (W3C) standards for

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access for disabled people and open access to web resources (for example, Cascading Style Sheets and HTML 4.01, and the W3C Web Access Initiative – see www.w3.org/WAI/). SoPSE persevered with a less than fully standardised Microsoft platform on the basis that solutions tended toward standards and future Microsoft compliance was perceived to be inevitable. The CMS solution selected bore this out.

Initially, we addressed a lack of common format documentation of local policy, formats, practices, selection criteria, target audiences, etc. We also researched technology, tools, and skills. Benefits of consortium working became apparent as we moved towards specification and procurement of complex software systems and defined clear responsibilities for content lifecycle stages.

We soon identified tasks the SoPSE consortium would address and those left to each partner. SoPSE partners are themselves partnerships, and some planned their own new collective infrastructure. SoPSE's focus was on delivering growing value to end-users and stakeholders without requiring massive and rapid changes to local practices, policies and procedures.

4. New management structures

The SoPSE Management Board and advisory Technical Group were established to examine standards and clarify and restructure financial arrangements, as well as plan shared contracting and technical delivery.

Staff were getting to know one another for the first time. Although common practice for research and development teams in industry and academia, many activities, requirements and emotions were novel for local authority staff unused to working across organisational boundaries. A constant issue was how to provide leadership without disempowering less experienced partners and thereby risking losing their effective participation. At each stage this meant a combination of open proposals, wide consultation, involvement with external projects, experts and stakeholders and intense work by very small teams and individuals working within agreed policy guidelines.

The end-user focus at SoPSE Board level required that the board acknowledge that the relationship between the sub-partners was not of interest in general to the board. This allowed the main (NOF-funded) partners to work at board level on collaborative features of the project alongside flexible and responsive partnerships focusing on development of content. This prevented arguments about digital resources while focussing attention on getting the most out of a shared technical resource.

We were building a sophisticated understanding of the knowledge management context within which the consortium will operate, at the same time as laying down pathways that will channel innovation and feedback.

All partners signed up to the vision and objectives of the project, and also to specific working arrangements. A Memorandum of Understanding acted as a formal contract to clarify the nature of commitments and responsibilities. This was a major landmark requiring partners to consult and gain permissions widely before signing.

The Technical Group drove site delivery activities from the start, but was always committed to the user. Its work centres on:

- establishing core (new) skills;
- developing internal standards, based on external and internal best practice;
- agreeing on key management principles but maintaining flexibility;

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- working together on common sub-tasks;
- devolving specialised tasks to experts;
- building on work already done;
- reporting to the board; and
- communicating.

Board decisions set working party tasks directly and minutes provided guidance. Recommendations to the board informed "work-tasks" in the business plans. The board was guided by feedback from the Technical Group. The Project Manager responsible reported regularly to the board on work done, requests for guidance and resolution of budgetary issues from the previous reporting period. Reports to the board, from all working groups, addressed both strategic and budgetary needs and took account of the original documentation in plans submitted in initial funding bids.

The Management Board discussed and agreed on critical decisions, and set priorities for key tasks within the risk analysis framework. It then monitored progress, continually evolving policy, and structured the budget and finance arrangements.

As the project moved on, we increased levels of involvement of project management, digitisation, design, learning resources and other specialist staff, extending the number of teams as required, but always working within the shared management structure to avoid duplication of effort and to promote effective communication.

5. Progress

It was clear at the end of 2001 that the main focus for the consortium Technical Group was the specification and purchase of the CMS. This is the software mechanism that co-ordinates the final web page presentation of material with the back-end processing required to deliver the actual archive material. The CMS assembles web pages and associated resources to meet the requirements of the end-user. The CMS needs to be a careful amalgam of database queries, templates, scripted pages and interactions with the actual archive data and images. Detailed design criteria for the software were drawn up for procurement. Research into appropriate systems and suppliers was carried out during the following year.

At the start of 2002 we had passed some critical milestones:

- employed a SoPSE Project Manager;
- published three "standards" papers to partners;
- established core working methods;
- employed partner project managers; and
- specified a web site in technical detail.

During 2002 we:

- Purchased a shared CMS, a customised version of the cross-platform "Index + " from System Simulation Ltd (www.ssl.co.uk/) following a public tender process and using technical advice provided through a contract with City University, London.

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- Engaged a graphic design company with extensive media and heritage experience, Illumina Digital (www.illumina.co.uk/), to work on all the sub-sites and the overall gateway, including navigation design.
- Finished research on the selection of suitable content materials.
- Started digitisation, in some cases after setting up new studio facilities.
- Completed a consultative process of web site graphic design that will provide distinct identities for each partner within a coherent overall design that will allow cross-site and intra-site search and browse facilities.
- Completed the initial technical implementation of the CMS, with our suppliers working on the web site.

During 2003, we secured contracts for our user interface design supplier and CMS supplier. They did not bid jointly to a single tender, as SoPSE wished to have the widest possible choice of suppliers for both aspects of the site. However, there was obviously a synergy in the way the two companies worked together and we benefited from their existing relationship.

We opted to use the graphic design company's web service to develop the design of the site in formal stages using the supplier's methodology, with full consultation amongst associate partners and quality assurance to ensure deadlines and expectations were both met.

The principle of building sustainable distributed digitisation capacity has meant that each main partner developed a digitisation and repository strategy for their own project, linked to the SoPSE CMS and delivery web service via common client-server software and shared metadata standards.

For example, the ThamesPilot consortium (www.thamespilot.org.uk) consists of 13 partners, but almost all photography and digitisation work was carried out by the ThamesPilot team in Maidenhead or by travelling to sub-partner sites, and metadata entry and data conversion work was shared between partners and the ThamesPilot team, supported by the SoPSE central team for final web delivery.

Since, there was no prior digital image server used for heritage materials, the ThamesPilot consortium set-up a high quality digital scanning unit and acquired Portfolio 6 and Adobe Photoshop to catalogue and batch process TIFF images for archival preservation and as digital masters for the JPEG files used for photographic image delivery. A link between the JPEG delivery images and the archival TIFF image and the metadata pertaining to the image content and rights and technical (Exchangeable Image File Format-EXIF) data is created by providing links within Portfolio-based metadata that is held in a central repository by ThamesPilot. This metadata, as well as the images, is exported and supplied to local sub-partners for their own use and to the central SoPSE web site for consortium public delivery.

In the case of Slough, microfilm of local newspapers was externally digitised and catalogued by the SoPSE partner unit before being added to the SoPSE server. Slough also set-up a ViewPortal Image Viewer server with shared funding, offering the whole SoPSE consortium the capability of delivering high quality, zoomable, web access to very large images. These range from Slough's broadsheet local newspaper pages to individual items in other partner collections, such as 12-foot wide maps of the Thames. SoPSE standards enable web site pages to offer direct links to images on this server

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without requiring it to be co-located with the main web server and CMS. Other specialised servers could be added in a similar way as they become available.

Other SoPSE partners were able to use existing museum catalogues or small databases by converting data to the SoPSE standard format, with metadata checked and, if necessary, expanded, by each partner's project team.

The standard format was based on the Dublin Core simple metadata element set and XML, extended as required with definitions and data formats. It also included a standard filename structure for all digital objects submitted for web publication. Filenames include codes for the contributing partner and local identifiers as well as object type and version information needed for batch loading and other processes. This makes it easy for both human recognition of unique files during editorial work as well as the sharing of component objects in consortium publications. The filename conventions are described in a document provided to consortium members, which describes the coding fields and separators used. The only slight downside is that filenames tend to be lengthy and do not conform to the MS Windows 8.3 convention used for older versions of the Microsoft operating system.

This centralisation has enabled rapid progress, but as other staff gain skills, such as digital photographic skills, these may replace or supplement temporary, expert project staff. The model is also highly flexible from a management point of view, since work can be carried out at various levels and sites, depending on finance and skills available and local needs. ThamesPilot, for example, set various annual fees (or no fee) for what it called associate, affiliate, supporting and linking partners. New local (or subject specialist) partners have been attracted by this formal offer, backed by the growing strength of the wider regional consortium, which is expected to provide affordable, high quality, sustainable delivery solutions that are widely promoted at local, regional and national/international levels.

One of the benefits of the SoPSE consortium has been that training in the use of the central Microsoft Windows-based CMS could be shared. The centralised project resources emphasised technology, but during 2004 focus shifted towards improving the overall user experience.

6. Lessons and experience for stakeholders

Partners have achieved a quality platform for publishing digital resources on SoPSE. They have also implemented their own digital repositories of archived materials.

Stakeholders of a project, such as SoPSE, include a wide range of groups. It is not merely the staff of the partner organisations, or even the audiences they plan to reach that should be counted. There are also government agencies, funding bodies, planners and managers of any community network development initiatives that might involve the use of cultural heritage resources, in addition to suppliers. Often the latter become partners in a technical sense. In the case of SoPSE, the resulting relationship transcends the vagaries of the current revenue stream.

SoPSE is one of a number of Sense of Place projects that cover almost the whole of the UK. Their potential as a long-term framework for further development and preservation of public digital resources is extremely significant. Examples include the Sense of the South West (www.sosw.org.uk/) Consortium, the EnrichUK (www.enrichuk.net/) site or search selected collections using the People's Network Discover (www.peoplesnetwork.gov.uk/discover/) service.

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Many issues have been raised by SoPSE in developing its web site and resources that have to incorporate not only the NOF-digitise Technical Advisory Service (2002) technical standards and guidelines, but also the needs of each of the various groups working in the consortium. Some of the main experience follows.

6.1 Management

We recognised the need for subsidiarity. Shared policy related to web-delivered resources, and the Technical Group did not seek to define common standards for preservation of masters.

A team approach was adopted spreading mainly top downwards. Team spirit and consultation at board level were facilitated mainly through regular (almost monthly to begin with) meetings, so that key strategic issues could be debated fairly and openly and policies set so that other teams could carry out detailed work independently of the board. Furthermore, we used common document frameworks and survey templates across all projects, so that information sharing was effective.

The Technical Group attracted high level representation with delegated powers to initiate and progress procurement. Collective communication with suppliers depended on the significant levels of trust that were built up. Expertise included local ICT staff, including PC, networking and database skills, project management, library and information technology research and development, risk management and professional museum, library and archive knowledge. Finance and local authority procurement expertise was provided at key stages.

In-depth background work supported the specification and procurement of systems. We adapted existing approved documents for suppliers rather than having to start from scratch.

E-mail discussions and web-based document sharing facilitated communication within the Technical Group and associate partners. It has, however, proved difficult to find a sustainable, low-cost, suitable host for project management resources, since the burden would fall more on one partner than the others and there are not yet any trusted external suppliers able to deliver targeted, low-cost services to the cultural heritage community in the region.

6.2 Human resources

We aimed for distributed information professional, artistic and pedagogical support, and to build a realistic option to create dedicated central posts when identified as necessary. The agreements also covered intellectual property rights (IPR), dissolution, high level working practices, principles and risk management.

Project managers had the benefit of outline workplans and detailed studies across partners to guide initial implementation. Successful pooling of expertise via special interest groups meeting regularly face-to-face and working via e-mail and the web was required. Using tools such as e-mail, intranets.com and MSN communities, we enabled consensus on allocation of responsibilities during procurements as partners became aware of skills, availability of individuals and organisational contexts.

We involved ICT specialists from several partners at an early stage, to assess local connectivity, security and firewall issues, and also to advise on hosting issues during and after procurement. It nevertheless remains a challenge to ensure that unproven

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procedures and practices relate to formal ICT planning processes in local government in a productive way.

6.3 Funding and sustainability

SoPSE partners continue to reflect on long-term methods of sustaining activity in a rapidly changing context of cultural heritage funding and regional development, recognised at events such as the sustainability conference in Bath (Lustie, 2004).

ThamesPilot has experimented with significant membership fees to provide some independence from external funding for shared staffing and services, with some success.

Other approaches being examined are the support of local publishers wishing to digitise and archive local newspapers; contact with tourism bodies to support provision of current information linked to location-based heritage resources; and the future extension of the consortium to members who might pay more for services than founder members.

Throughout, significant sums have been committed to shared costs. These include funds for project management, consultancy, CMS and design procurement. We have also maintained an ongoing discussion on the cost-benefits of centralised versus separate systems for other functions.

We have perhaps not so much made savings, but strengthened procurement, server bandwidth, and systems functionality that will enhance future service delivery. However, there have been additional co-ordination costs, particularly at the set-up stage of the consortium, that probably balance or outweigh any short-term financial benefits.

A long-term planning horizon created by the consortium enables strong stakeholder involvement and the will to resolve difficulties. The ThamesPilot consortium has itself employed a specialist fundraising expert who is working with partners to develop an additional and completely new formal partnership that will better suit a funding balance of government, local government and charitable sector funding.

6.4 Productivity and impact

It is still too early to assess the overall effect of the consortium and levels of productivity achieved. Not all partners have progressed at a similar rate. Their size, budget, rights problems, local contexts and technical issues have affected innovation.

By having a ready-made resource catalogue, Reading Museum was able to complete its Huntley and Palmers web site (www.huntleyandpalmers.org.uk) relatively swiftly, as a kind of digital exhibition activity, although the partner remains actively involved in consortium management and highly committed to further future co-operation as opportunities arise. Hantsphere faced technical difficulties, wishing to include fully digitised books as content, requiring more complex search functionality than that required for images. Productivity was certainly greatest when cohesive teams were in place.

Managers from the start considered it important to retain the flexibility to consider a future independent trust status for the consortium, or a return to separate servers per partner linked only by standards, practices and service agreements.

These decisions have affected procurement. They have also been made on the basis that it is essential to build trust between partners. We needed to give coherent technical advice to suppliers in addition to providing effective operational support in a short timescale.

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Consortium solutions, although they complicate design initially, mean local partners should find it easier to work as providers to other consortia or channels in future, since a layered approach has been taken, for example to preservation, authoring, production and web delivery.

Some partners may wish, or be forced by local factors, to reduce their future level of support for the consortium. However, the regular reports back to the board, the continued contributions to working parties and the need to communicate regularly on standards and operations, provide a platform of continued co-operation. This closeness means the board is provided with the maximum of advanced warning of problems arising with either funding or commitment from one or more of the consortium partners. There are also opportunities for new members to join and it is hoped that this will strengthen the alliance.

6.5 Priorities

The user is central to the consortium's vision. Involving staff and end-users in digital service developments is not yet widespread, but changes are already significant for the project managers and other key staff, and already several new types of post have been created that are likely to evolve and expand as services develop.

The inclusive development of an information architecture model will allow for the secure creation and preservation of digital resources and, in particular, the final delivery of a public archive of digital resources. This will be compliant with NOF-digitise mandatory and recommended standards, but more importantly will meet local community needs that are constantly evolving.

6.6 Technical aspects

A key success is the solid, shared, technical platform for service delivery, suitable for use by both small and large partners in a flexible manner. This is based on a shared resource-naming scheme based on hierarchical elements that enable each resource file to be quickly identified as belonging to a particular partner/sub-partner in the overall SoPSE consortium. Local, and SoPSE partner, unique identifiers enable structured linking, resolution of human queries, version control, local preservation and other management tasks to be carried out. This scheme, subsequently known as the SoPSE Filename Schema, resulted from some initial research based on a meeting of the two authors of this paper. Then, a scheme was compiled based on over eight years' experience of file handling in databases and operational web site management. The schema was subject to a document control procedure and began a long series of minor and major revisions to ensure that it met the rigorous needs of the organisation. As we write this paper, late in 2005, the scheme is in version 19 of its document revision history. It is still based on the original one, but the changes and revisions mean that it is now almost perfectly aligned with our management of the resources held across the consortium (further details available from Damon Guy).

We have common access points and some terminologies, although it has not been practical to apply existing standards or tools off-the-shelf. Our use of shared templates has reduced costs and spread awareness of opportunities. These offer searching, navigation functionality, layout, and access to help.

There have been collective attempts at high volume digitisation, agreement on taxonomies, and preservation strategy. No partner had existing strong solutions and all have redefined their own strategies in line with partners.

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Significant technical challenges remain, particularly within what is perceived as a sustainable budget. An example of this is work for Slough newspapers, where delivery of potentially large, commercial, image files is still a problem, and one that will be faced by many more local newspaper projects. However, as costs come down, there is now a shared framework for exploring and implementing new solutions. New ideas on repositories, rights management and interoperable authentication remain to be dealt with by the consortium partners.

6.7 Problems encountered

Throughout the project term and since the end of the NOF funding, the SoPSE project team have of course encountered a range of problems. We outline some of these here.

6.7.1 Insufficient funds. The formation of the consortium led to some cost savings (i.e. the savings from shared CMS and public facing servers). On the other hand, it has led to some additional costs. These are not so obvious: higher travelling costs (between meetings, training, etc.), a requirement for a project manager post (for co-ordination) and higher levels of technical input/expertise than would have been required if each project had acted separately. NOF recognised the additional administration burdens and assisted with extra funding. But the additional hidden costs from scaling up the technology for a larger operation were not given assistance.

Calculations and tests showed that SoPSE would need one terabyte of server space for the public facing server. One project (Slough) needed that amount of server space for their own archive of TIFF format images.

On award of funding the cost of required server space seemed out of reach. To ensure the continuation of the programme, Slough elected to be the last project online. This allowed us to capitalise on the rapidly falling cost of hard drive space. It took nearly three years, however, before the funds stretched to the purchase of appropriate image server technology. This led to severe operational problems at later stages.

The delay meant uneven progress across the overall SoPSE project implementation. Projects went online one after another, but funding started at the same point. Thus, at the end of the funding period, the last project comes online – when the first project online was running out of set-up funds. The hosting overall was also by then running out of funding. This problem, still unresolved, has magnified the need to find new ways to sustain the overall SoPSE project now external funding has terminated.

6.7.2 Intellectual property rights. Probably nearly all NOF-digitise projects found IPR issues to be more of a challenge than expected. NOF had asked for the right to repurpose project images in their own marketing and promotion. This led some projects to be in extended negotiations with the owners of images and in some cases material had to be withdrawn. Repurposing of images is a very problematic issue in IPR contexts. Other legal issues around putting images online loomed large in the minds of owners which lead to protracted decision times or undue legal costs (not supported by NOF).

Some of the determination to ensure that SoPSE survived came out of the early period when each project was reporting to the board the success and failure of various IPR negotiations. The clarity of purpose for our Memorandum of Understanding arose from the same meetings where board members were reporting IPR difficulties. The more we seemed to find it difficult to get our partners to agree, the more the board seemed to crystallise around a single purpose.

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One key conclusion is that projects should allow sufficient time and resources to deal with these IPR issues. It is also clear that local authority legal staff are not necessarily equipped to deal with IPR!

6.7.3 Working separately, operating together. The federated structure of SoPSE has led to flexibility and responsiveness to the needs of partners and site users. However, local staff were not working so close to each other as board and working party members. Two issues could be improved in future projects:

(1) *Training*. All the projects were working to the same standards set by NOF and using the same content management technology. However, there was insufficient sharing of knowledge. Good manuals and shared training sessions have helped to offset this effect. However, as the project gets older and some of the "old guard" leave, the newer staff doing the day to day work are not sharing the same commitment to the overall project. We are, therefore, working on developing shared activities which contribute to shared commitment and help focus the purpose of staying together.

(2) *Implementing standards*. Diminished communications also reduce standardisation benefits. Projects try to make small local adjustments that threaten the integrity of the system. While we maintain aspects of the scheme by consensus at board level, it is not easy to ensure that this is rigorously applied to every record entered into our databases at the front-end of each project. In the event of a disaster or if a large-scale operational change is required in future, the result of lots of small variations in the standards by each project will lead to unforeseen costs in scripting and file management. The integrity problem is well known in database circles, but perhaps that lesson is looming for many projects which have been involved in this "consortium" model. The probable resolution is a greater degree of working together by the various project groups. This does carry high costs however.

7. Towards a collaborative model

We confirm that developing sustainable support for cross-domain digitisation projects involving the public sector presents a large number of diverse challenges. The SoPSE project started with a group of dissimilar collections with little obvious commonality except a feeling of a "sense of place".

Although this focus is shared with several other NOF-digitise consortia and many other local projects, it has yet to be demonstrated how the territorial approach relates to topical themes both within the consortium and beyond, and what benefits it can bring to users. Local accountability and sustainability may be key to this, but also the provision of unique resources to the public digital commons.

If people use Google for discovery, perhaps a national strategy for public heritage web services is required, which may be built by extending existing consortia. SoPSE has already taken the initiative to make available Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH) based interoperability. New intellectual linkages can then be made by the community at large as well as information professionals.

Future areas of interest include:

- How should SoPSE services develop, now that lottery funding has apparently ceased?

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- Which audiences benefit most from digital services, and what are their future needs?
- What management methods work best to build and spread the new staff skills that are required? Are there specific communities of practice that should be developed?
- How should the limited content management plans developed for SoPSE be related to external developments within partners and beyond?
- What steps need to be taken to ensure a robust future technical infrastructure is provided at an appropriate cost?
- What mix of centralised and distributed service support is appropriate for digital service provision?
- How can SoPSE help to support the local economy?
- Does SoPSE help promote social cohesion, joined-up thinking and a sense of community?
- What outcomes are there that need to be disseminated and how should this be done most effectively?

Answering these questions will require continued sharing of information, self-assessment and leadership, as well as further training and development of staff. We shall need to compile an evidence bank to support assessment, taking into account the various stakeholder perspectives.

8. Conclusions

The SoPSE digital cultural heritage consortium clarifies the roles of local authorities, suppliers and content holder communities as well as other stakeholders. It is bringing commercial technical expertise, based on international research and development, together with distributed, local, operational teams that are building new skills, despite operational challenges. It is also clarifying the economic and social context for digital services, and strengthening the basis for future community support at various levels. We integrate at national level with the People's Network, at regional level with the South East Museum Library and Archive Council (the regional development agency for the museum, library and archive sector in the South East), at local authority level with the principal content providers and at professional level, through the appointment of key, experienced, dedicated individuals. We can, therefore, effectively plan what we hope will be sustainable services that meet genuine needs and are attractive to audiences. Our communities will not only be able to enjoy new dimensions of their sense of place, but we shall be able to demonstrate the cultural value of such collaboration to our many stakeholders.

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Corresponding author

Robin Yeates can be contacted at: robin.yeates@barnet.gov.uk

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